



UPR
Recinto Universitario de Mayagüez

Estimation of Change in Population of Municipalities in PR

[ESMA 6835]: Bayesian Inference

Alejandro M. Ouslan & Tatiana Crespo

University of Puerto Rico, Mayagüez

March 5, 2026

Abstract

This research aims to compare the performance of spatial regression models that rely on predefined weight matrices with that of semi-parametric regression models using spatial smoothers. This research will be using

Contents

1	Introduction	3
2	Methodology	3
3	Resultados	3
3.1	Densities	3
3.2	Credibility Interval	3
3.3	Confidence Interval	3
3.4	Distribution	3
4	Predicciones	4
5	Conclusions	4

1 Introduction

Change in demographics are highly studied topic and is of great interest to governments. Knowing the demographics characteristics are of special interest to regional governments like the municipalities in Puerto rico. Understanding the how the population will change in the following years are the basis of many infra stare projects to invest in a growing population or a population that is decreasing or aging.

2 Methodology

This research will be using data form the American Community Survey 5 estimates (ACS) by the U.S Census Bureau. This data encompasses population estimates from 2011 to 2023 for all 78 municipalities.

3 Resultados

- Regional data often exhibit spatial dependence.
- Example: Puerto Rico municipalities.
- Productivity in one region may depend on neighboring regions.
- Violates classical i.i.d. assumption.

3.1 Densities

- Regional data often exhibit spatial dependence.
- Example: Puerto Rico municipalities.
- Productivity in one region may depend on neighboring regions.
- Violates classical i.i.d. assumption.

3.2 Credibility Interval

- Regional data often exhibit spatial dependence.
- Example: Puerto Rico municipalities.
- Productivity in one region may depend on neighboring regions.
- Violates classical i.i.d. assumption.

3.3 Confidence Interval

- Regional data often exhibit spatial dependence.
- Example: Puerto Rico municipalities.
- Productivity in one region may depend on neighboring regions.
- Violates classical i.i.d. assumption.

3.4 Distribution

- Regional data often exhibit spatial dependence.
- Example: Puerto Rico municipalities.
- Productivity in one region may depend on neighboring regions.
- Violates classical i.i.d. assumption.

4 Predicciones

Due to the lack of significance in the results of the change in time and municipality population, no mining full recommendation can be made on the problem.

5 Conclusions

The overall results of the simple regression are inconclusive given that year was not significant. This lack of statistical significance may be due to the limitations of a simple regression. A major component of this limitation is the violation of the assumption of i.i.d. Population are highly correlated to the population of the previous year. It is advised to use a more appropriate model for time series like for example ARIMA.